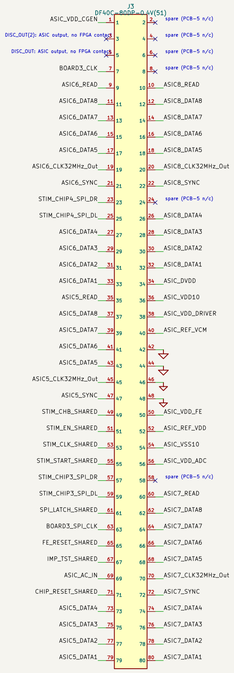
FPGA–LDO PCB V33 VERSION B – ROUTER (STAYS ON THE HEAD)

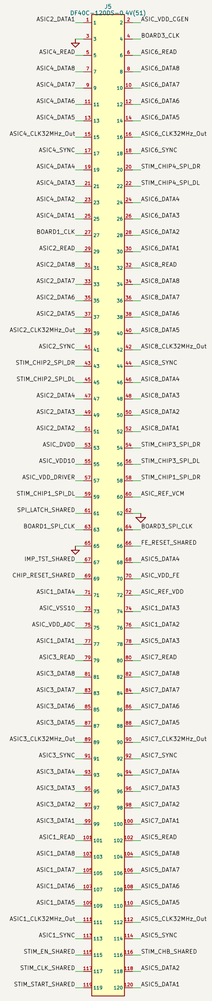
ASIC PORTS – TWO HELENA PCB-5 PAIRS (J3 TOP, J4 BOTTOM, SAME SPOT)



1. Function: Two PCB-5 pairs sandwich this board. J3 (plug, top face) feeds board 3 (near: ASIC5/ASIC6) and board 4 (far, via T_ lines: ASIC7/ASIC8). J4 (receptacle, bottom face, same XY) feeds board 1 (near: ASIC1/ASIC2) and board 2 (far: ASIC3/ASIC4). Each port also carries the nine ASIC rails. AC_IN. 2 AGND and 2 DGND contacts.

2. Source / changes: ASIC board PCB-5 (Reference/ASIC5 PCB 1K/PCB – Bottom.zip, PCB.kicad_pcb): J3 DF40C-80DS on B.Cu and J4 DF40C-80DP on F.Cu carry the same net on each pin number, so every contact is bused up the stack; T_ nets are through-only and feed the second (far) board. Claude pin assignment (as V17): D<kx>=DATAk, chip 1 = odd ASIC, chip (2) = even ASIC; the far board shares CLKH, SPI_CLK, SPI_DINL/R and the stim lines with the near board; DISC_OUT and spares 4/6/8/24/58 open. Unverified: far-board wiring, recording-only boards sharing stim-chip SPI data.

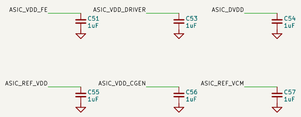
LDO–BOARD LINK – J5 DF40C–120



1. Function: J5 on the router's underside, south of J4, plugs onto the LDO board. pin n to pin n, 1.5 mm mated. It carries the 100 port signals, the B shared controls, the nine ASIC rails, 1 AGND and 2 GND: exactly the version A port nets except AC_IN, whose default (R6/TP7) sits on the router. So B = A split at J5.

2. Source / changes: Claude design (Howard chose one DF40C-120 on 7 Nov 2026). Hirose drawing EDC3-311351-00 DF40C-*DP-0.4V(51), rev 2 (2024-07-08. HRS Drawing for Reference, printed 1 Oct 2026), note 2: the four corner metal fittings have no electrical continuity; they are separate pads 0.475 mm beyond the contact rows (KiCad MP pads), so every numbered contact is live. DF40C-120DP 25.52 x 2.97 mm (same drawing, table): 0.3 A per contact and 30 mating cycles (Hirose DF40C-120DP product page CL0684-4107-9-51). Stock KiCad 10 footprints (120 contacts + 4 MP fittings on the header). Pin order: J5 pin map V4 (10 Nov 2026, version B floorplan V1): column 1 west, odd pins = north row; J1 nets on columns 2–26, J2 nets on 27–60, rails straight under their J3/J4 pins; lanes checked on signals-only boards, not a layout. Unverified; signal return with only 2 GND + 1 AGND contacts for about 100 signals: rail current per contact (a few mA per rail).

RAIL CAPACITORS AT THE ASIC PORTS



1. Function: C51 and C53–C57 put 1 uF on six ASIC rails right at the J3/J4 ports, so each has local charge where it enters the PCB-5 stacks; all return to AGND.

2. Source / changes: Claude design (7 Nov review, V20), no manufacturer reference circuit. V28 (Howard, 9 Nov 2026): C52 (VDD_ADC), C58 (VDD10) and C59 (VSS10) removed – no clean layout spot on their rails; VDD_ADC keeps U3's 1 uF C302 beside the pins. VDD10/VSS10 keep 22 uF + 10 uF at U9/U10. V23: C52 (VDD_ADC) moved from GND to AGND to match Gerald's LDO Board A, whose VDD_ADC load decaps C31/C32/C38/C39 go to AGND while its regulator U2 and C8 sit on DGND (as U3/C302 here), and PCB-5 C1/C34 (AGND). In version B the rails reach the router through one J5 contact each and then cross J3/J4, so these parts sit on the router; version A has them too, so A and B stay the same circuit; All are 10 V 0402. Unverified: decoupling already on PCB-5, and the regulators' stability with this extra capacitance at the end of the trace (LT3042/LT3094 are stable with larger output capacitance per their data sheets).

AC_IN DEFAULT



1. Function: R6 holds the analog AC_IN input of all eight ASICs at AGND; TP7 lets a bench source inject a test signal (0–1.5 V nominal) through the shared node.

2. Source / changes: Claude design, The Frontend has no AC_IN source and no connector contact for it (V43), so AC_IN is handled here. V21: in version B these parts sit on the router, so AC_IN needs no J5 contact. Nominal 0–1.5 V from ASIC_REFERENCE.md section 4 (relayed designer answer). Gerald, 9 Nov 2026: "0V is the right state. No need to care about input impedance or anything else."

Returns arrive from the LDO board (NT1 bond there)

ASIC ports J3/J4: J5 DF40C-120 underneath to the LDO board.
No regulators: rails and returns come from the LDO board.
Schematic only, no layout. Input: VIN_3V3 from Frontend J2.68–70.

Rice – FPGA PCB workspace

Sheet: /
File: FPGA_LDO_V33B_Router.kicad_sch

Title: FPGA–LDO V33 B – router (on the head)

Size: User	Date: 2026-11-10	Rev: V33
KiCad E.D.A. 10.0.6		Id: 1/1